

# Parbhat Kapila

AI Systems Engineer

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## Summary

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Product engineer who **owns every layer**: complete AI systems shipped solo in TypeScript, from RAG retrieval and autonomous agents to a 12-stage video pipeline, each with its own auth, billing, queues, rate limiting, and observability. Design principle: restraint. Confidence gates before irreversible actions, grounded answers with citations, cost budgets in the request path, fail-closed defaults.

## Professional Experience

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### AI Systems Engineer, Independent (parbhat.dev)

Remote

*AI Application Development & Cloud Infrastructure*

*May 2022 – Present*

- Shipped **5 AI products** solo, all live today (Sentinel, CUTLINE, RepoDoc, VectorMail, Visura), TypeScript throughout: Next.js + tRPC frontends, Node services, Postgres + pgvector, Redis + BullMQ.
- Owns **the production surface most AI demos skip**: OAuth across Google, Slack, HubSpot, and Salesforce; billing with plan entitlements enforced at the UI, API, and database layers; rate limiting and RBAC.
- Failure is designed for, not patched**: atomic job claiming with leases, idempotency keys on every side effect, provider fallback chains; zero-downtime deploys on AWS and Vercel, Sentry + OpenTelemetry tracing on every system.

## Key Projects

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### VectorMail: Autonomous Email Agent [GitHub](#)

- Autonomous email agent with a hard **read rail / write rail separation**: semantic search with cited source messages on one side; drafting and sending, double-gated by a topic gate plus outbound moderation, on the other.
- The model sits upstream of the gate, **never trusted to be the gate**: automation runs dry-run by default and every action carries a unique idempotency key, so double-execution surfaces as a database error instead of a duplicate send.
- Everything lives in **one Postgres**: embeddings as vector(768) columns on the email row, tenancy enforced in the same WHERE clause as retrieval; incremental Gmail history sync; calendar booking via Aurinko.

### Sentinel: Revenue Pipeline Intelligence [GitHub](#)

- Treats the **CRM as the permission layer**: ingestion from HubSpot, Salesforce, Gmail, and Slack is gated on CRM contact matches and fails closed, so each rep sees only deals they're authorized on, zero manual access config.
- Time-decay scoring, stage velocity, and engagement signals **flag at-risk deals before they quietly rot**.
- Real-time SSE alerts, Redis-backed rate limiting, and RBAC **replace manual pipeline review**, so reps get notified automatically instead of waiting for a manager to spot the problem.

### CUTLINE: Automated AI Video Pipeline [GitHub](#)

- One sentence in, one finished MP4 out: a **12-stage deterministic pipeline** with no templates and no human handoffs, wiring BullMQ, Remotion, and ElevenLabs into one job graph.
- Per-stage failure isolation**: each stage retries independently and streams real-time progress to the frontend; a 4-provider image fallback chain means a render never dies on a missing asset.
- Multi-model routing** via OpenRouter keeps a slow or failed LLM from ever blocking the render queue; provider content-filter blocks trigger an automatic narration reword that preserves visuals and re-syncs captions.

### RepoDoc: Codebase RAG Infrastructure [GitHub](#)

- Embeds **what each file means, not what it says**: every file is LLM-summarized and the summary embedded; retrieval is sub-second pgvector cosine similarity, and every answer carries exact file + line citations engineers can verify.
- Postgres is the job queue**: indexing jobs claimed by atomic compare-and-swap with 5-minute leases and cursorized resume, so indexing survives serverless timeouts without a dedicated worker.
- Cost control in the hot path**: per-project monthly budgets return 402 and pause in-flight indexing mid-job; a pre-index path answers from ~22 high-value files fetched live, so the tool works before indexing finishes.

## Technical Skills

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**Languages:** TypeScript, SQL

**Frontend:** React, Next.js, Tailwind CSS

**Backend:** Node.js, Express, tRPC, REST APIs, WebSockets, SSE, BullMQ

**Databases & Caching:** PostgreSQL, pgvector (HNSW), Prisma, Drizzle, Redis, Supabase, Neon

**AI Systems:** RAG pipelines, vector embeddings, LLM orchestration, multi-model routing (OpenRouter), OpenAI API, LangChain.js, ElevenLabs, Remotion

**Auth & Billing:** OAuth 2.0 (Google, Slack, HubSpot, Salesforce), Clerk, Better Auth, plan entitlements, webhook idempotency

**Cloud & Observability:** AWS (EC2, S3, Lambda), Vercel, Docker, GitHub Actions, CI/CD, Sentry, OpenTelemetry, structured logging, Vitest